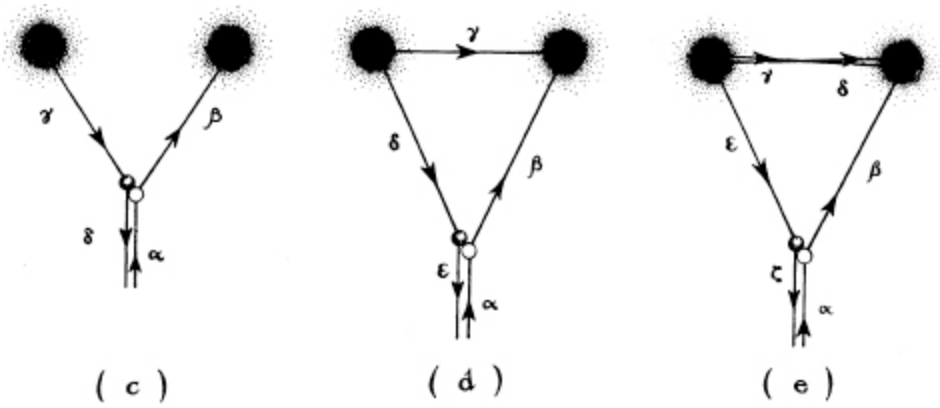


Some months later, Robert Forward [one of the pioneers of laser interferometer gravitational-wave detectors (Chapter 10) and also a science fiction writer] discovered a third trajectory that satisfies all the laws of physics, the trajectory (c) below: The collision, instead of occurring between the mouths, occurs before the ball reaches the mouths' vicinity. I then realized that the collision could be made to occur earlier and earlier, as in (d) and (e), if the ball travels through the wormhole several times between its two visits to the collision event. For example, in (e), the ball travels up route α, gets hit by its older self and knocked along β and into the right mouth; it then travels through the wormhole (and backward in time), emerging from the left mouth on γ, which takes it through the wormhole again (and still farther back in time), emerging along δ, which takes it through the wormhole yet again (and even farther back in time), emerging along ε, which takes it to the collision event, from which it is deflected down ζ.



Evidently, there are an infinite number of trajectories (each with a different number of wormhole traversals) that all satisfy the classical (non-quantum) laws of physics, and all begin with identically the same initial conditions (the same initial billiard ball location and velocity). One is left wondering whether physics has gone crazy, or whether, instead, the laws of physics can somehow tell us which trajectory the ball ought to take.